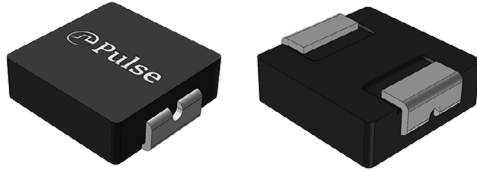


SMT Power Inductor

High Current Molded Power Inductor - PA5402 & PM5402 Series



- Ⓟ **Height: 2.2mm Max**
- Ⓟ **Footprint: 4.7mm x 4.31mm Max**
- Ⓟ **Current Rating: up to 12A**
- Ⓟ **Inductance Range: 0.1 to 22uH**
- Ⓟ **High current, low DCR, and high efficiency**
- Ⓟ **Shielded construction and compact design**
- Ⓟ **Minimized acoustic noise and minimized leakage flux noise**
- Ⓟ **200 Vdc Isolation Between Terminal and Core**
- Ⓟ **Available in Commercial (PA) and Automotive (PM) grades**

Electrical Specifications @ 25°C - Operating Temperature -55°C to +125°C

Commercial ^{6,7}	Automotive ^{6,7}	Inductance ⁵ 100KHz, 1.0V	Rated ³ Current	DC Resistance		Saturation ² Current
			TYP.	TYP.	MAX.	TYP.
		uH±20%	A	mΩ	mΩ	A
PA5402.101NLT	PM5402.101NLT	0.10*	12	3.2	4	35
PA5402.181NLT	PM5402.181NLT	0.18*	13.5	4.6	5.4	28
PA5402.221NLT	PM5402.221NLT	0.22*	13	6.6	7.3	24
PA5402.331NLT	PM5402.331NLT	0.33	10	7.8	8.6	18
PA5402.471NLT	PM5402.471NLT	0.47	8	11.2	14	12
PA5402.561NLT	PM5402.561NLT	0.56	7.3	13.5	16	10
PA5402.681NLT	PM5402.681NLT	0.68	7	16	19	10
PA5402.102NLT	PM5402.102NLT	1.0	5	22	27	8.5
PA5402.122NLT	PM5402.122NLT	1.2	4.8	25	30	7.8
PA5402.152NLT	PM5402.152NLT	1.5	4.5	34.8	42	7
PA5402.222NLT	PM5402.222NLT	2.2	4	51	61	6
PA5402.332NLT	PM5402.332NLT	3.3	3.5	69	76	4
PA5402.472NLT	PM5402.472NLT	4.7	2.6	95	105	3.5
PA5402.562NLT	PM5402.562NLT	5.6	2.2	112	125	3
PA5402.682NLT	PM5402.682NLT	6.8	2.1	150	172	2.8
PA5402.822NLT	PM5402.822NLT	8.2	2	158	180	2.5
PA5402.103NLT	PM5402.103NLT	10	1.8	215	243	2.3
PA5402.153NLT	PM5402.153NLT	15	1.5	325	374	1.9
PA5402.223NLT	PM5402.223NLT	22	1.2	470	500	1.4

SMT Power Inductor

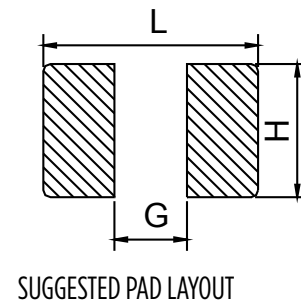
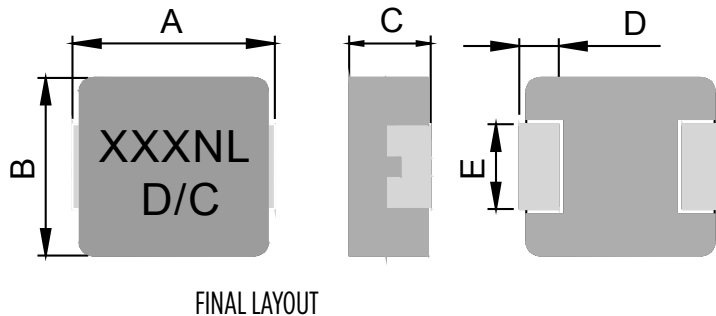
High Current Molded Power Inductor - PA5402 & PM5402 Series

Notes:

1. Actual temperature of the component during system operation (ambient plus temperature rise) must be within the standard operating range.
2. The saturation current is the current at which the initial inductance drops approximately 30% at the stated ambient temperature. This current is determined by placing the component in the specified ambient environment and applying a short duration pulse current (to eliminate self-heating effect) to the component.
3. The rated current is the DC current required to raise the component temperature by approximately 40°C. Take note that the components' performance varies depending on the system condition. It is suggested that the component be tested at the system level, to verify the temperature rise of the component during system operation.
4. The part temperature (ambient+temp rise) should not exceed 125°C under worst case operating conditions. Circuit design, PCB trace size and thickness, airflow and other cooling provisions all affect the part temperature. Part temperature should be verified in the end application.
5. Please note that the inductance tolerance of all parts are $\pm 20\%$, except those indicated by an * which are $\pm 30\%$.
6. Parts shown in bold are standard catalog parts and are available through sample stock and distribution. Parts in lighter font are available but are not necessarily held in sample stock or distribution and lead times may be longer. Please contact Pulse for availability.
7. The PM prefix parts are AEC-Q200 qualified and has full automotive IATF16949 certification. The mechanical dimensions are 100% tested in production but do not necessarily meet a process capability index (Cpk) 1.33 and therefore may not strictly conform to PPAP.
8. Special characteristics

Mechanical

PA5402/PM5402



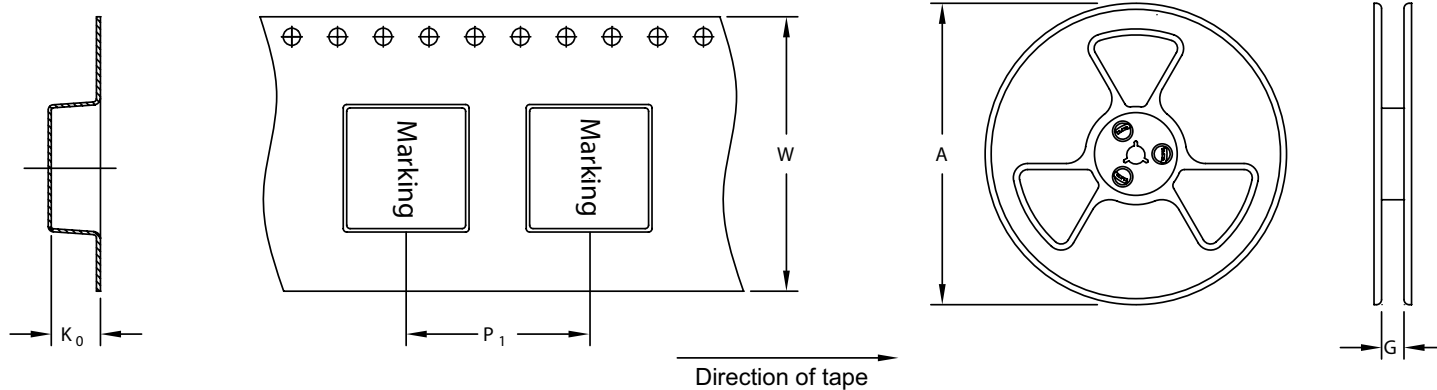
Series	A	B	C	D	E	L	G	H
PA5402/PM5402	4.45 $\pm$ 0.25	4.06 $\pm$ 0.25	1.8 $\pm$ 0.2	0.76 $\pm$ 0.3	2.0 $\pm$ 0.2	5.2	2.2	2.4

All Dimensions in mm.

SMT Power Inductor

High Current Molded Power Inductor - PA5402 & PM5402 Series

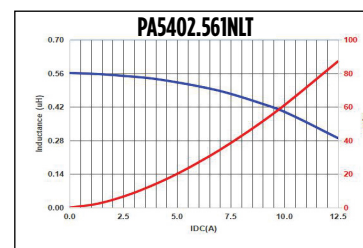
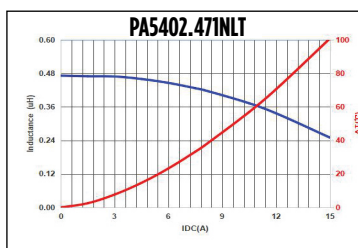
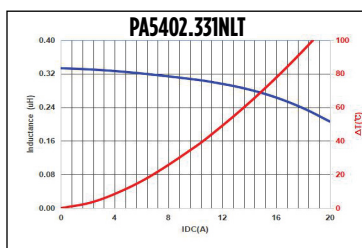
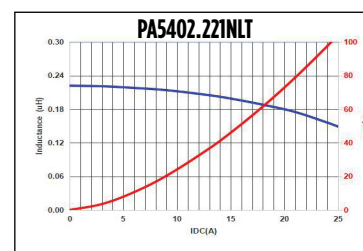
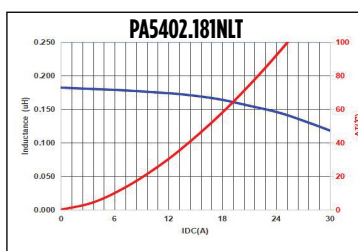
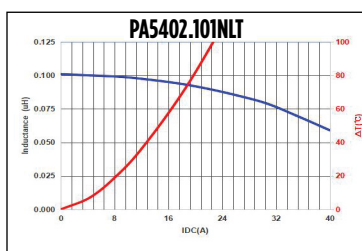
TAPE & REEL INFO



SURFACE MOUNTING TYPE, REEL/TAPE LIST						
	REEL SIZE (mm)		TAPE SIZE (mm)			QTY
	A	G	P_1	W	K_0	PCS/REEL
PA5402/PM5402	Ø330	12.4	8	12	2.3	3000

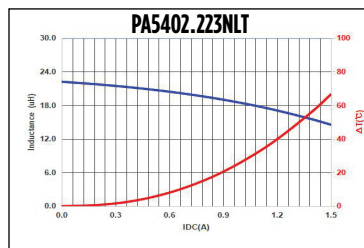
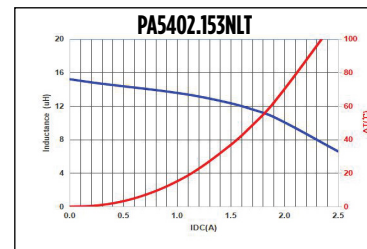
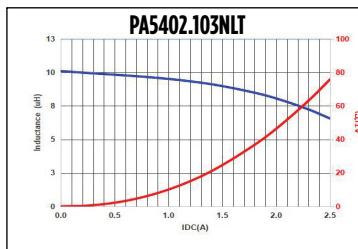
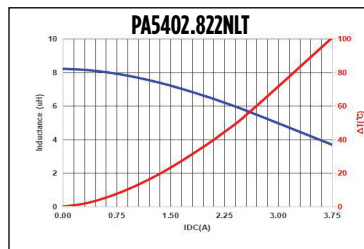
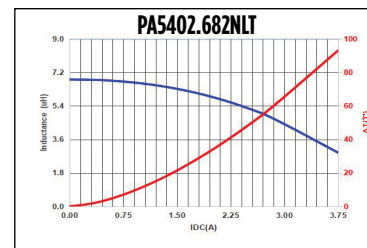
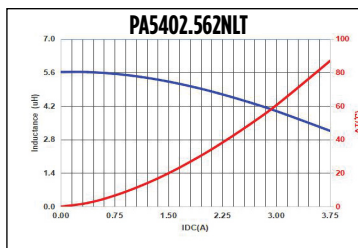
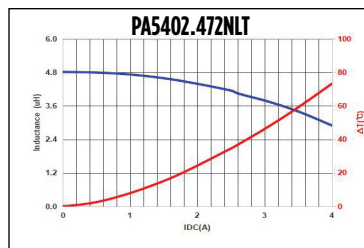
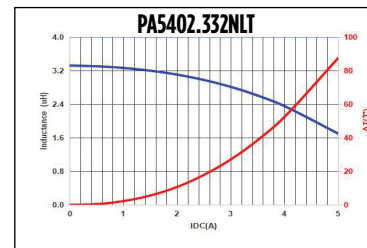
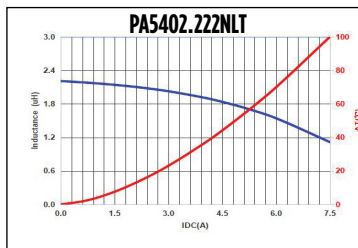
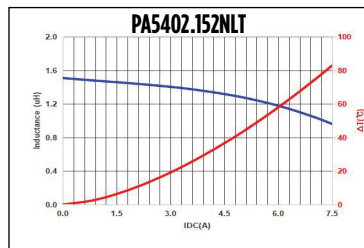
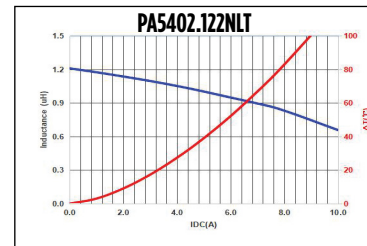
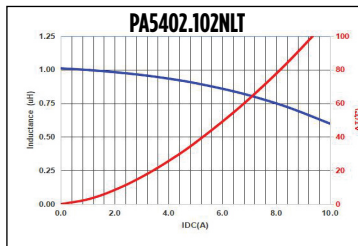
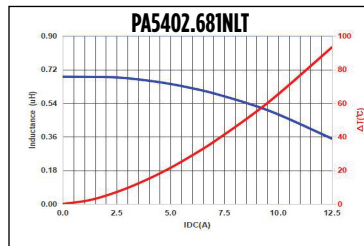
Typical Performance Curves

PA5402/PM5402



SMT Power Inductor

High Current Molded Power Inductor - PA5402 & PM5402 Series



For More Information:

Americas - prodinfo_power@pulseelectronics.com | Europe - power-apps-europe@pulseelectronics.com | Asia - power-apps-asia@pulseelectronics.com

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Mouser Electronics

Authorized Distributor

Click to View Pricing, Inventory, Delivery & Lifecycle Information:

Pulse:

PA5402.101NLT	PA5402.102NLT	PA5402.103NLT	PA5402.122NLT	PA5402.152NLT	PA5402.153NLT
PM5402.682NLT	PM5402.822NLT	PM5402.332NLT	PM5402.471NLT	PM5402.472NLT	PM5402.561NLT
PM5402.562NLT	PM5402.681NLT	PM5402.153NLT	PM5402.181NLT	PM5402.221NLT	PM5402.222NLT
PM5402.223NLT	PM5402.331NLT	PA5402.822NLT	PM5402.101NLT	PM5402.102NLT	PM5402.103NLT
PM5402.122NLT	PM5402.152NLT	PA5402.471NLT	PA5402.472NLT	PA5402.561NLT	PA5402.562NLT
PA5402.681NLT	PA5402.682NLT	PA5402.181NLT	PA5402.221NLT	PA5402.222NLT	PA5402.223NLT
PA5402.331NLT	PA5402.332NLT				